

NAKISKA RIDGETOP, AB, Canada

WMO: 712450

Lat: 50.9437N

Lon: 115.1896W

Elev: 2543

StdP: 74.28

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.9	-24.0	-32.5	0.2	-22.8	-28.9	0.4	-18.6	32.7	-5.1	27.7	-5.8	3.6	50	1.058

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	18.4	8.9	16.8	8.3	15.3	7.7	10.6	15.4	9.6	14.6	8.6	13.9	5.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	9.7	11.7	7.4	8.7	11.0	6.1	8.0	10.2	38.2	15.5	35.2	14.7	32.6	14.0	15.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.4	23.6	19.4	DB	-32.0	21.5	3.7	1.9	-34.7	22.8	-36.8	23.9	-38.9	24.9	-41.6	26.3	(4)
(5)			WB	-31.3	12.4	4.0	1.4	-34.2	13.4	-36.5	14.2	-38.8	15.0	-41.7	16.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-8.8	-9.8	-7.5	-3.8	1.3	4.8	9.5	8.9	4.5	-1.7	-6.6	-9.3
	DBStd	8.51	5.98	5.91	5.29	4.59	4.44	3.81	4.08	4.07	5.28	5.04	5.51	5.73
	HDD10.0	4298	584	555	544	414	272	162	59	71	178	362	499	599
	HDD18.3	7237	842	788	802	664	529	404	274	293	414	619	749	857
	CDD10.0	102	0	0	0	0	1	7	44	36	14	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.3	9.7	7.1	8.6	6.5	5.9	5.6	5.8	5.3	6.6	7.9	9.6	9.3
Precipitation	PrecAvg	654	53	36	44	39	60	89	62	58	54	50	59	50
	PrecMax	857	83	57	76	84	105	201	118	111	103	94	138	79
	PrecMin	509	16	13	13	18	28	53	25	16	12	27	26	27
	PrecStd	69	14	12	16	14	19	33	23	23	23	16	25	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	2.8	5.3	9.8	13.9	17.8	21.2	20.2	18.2	12.6	6.5	5.4
		MCWB	-2.3	-2.7	-1.5	2.8	5.5	8.5	10.3	9.4	7.6	4.3	0.2	-1.6
	2%	DB	1.8	0.9	2.0	6.7	11.6	14.8	19.0	17.9	15.8	9.4	4.1	2.3
		MCWB	-3.6	-3.2	-2.4	1.1	4.4	7.7	9.5	8.8	7.0	2.3	-0.6	-3.0
	5%	DB	-0.2	-1.1	0.2	4.6	9.7	12.6	17.2	16.3	14.0	7.3	2.3	-0.1
		MCWB	-3.8	-4.1	-2.7	0.0	3.5	6.6	8.8	8.1	6.2	1.7	-1.0	-3.7
	10%	DB	-1.8	-2.7	-1.4	2.7	7.9	10.5	15.6	14.8	12.2	5.5	0.4	-2.0
		MCWB	-4.4	-4.9	-3.7	-0.9	2.6	5.3	8.1	7.6	5.4	0.8	-2.4	-4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	-1.2	-0.2	3.1	6.6	10.7	12.7	11.6	9.5	5.5	1.7	-0.5
		MCDB	3.8	1.4	3.3	8.5	12.5	14.7	17.9	15.5	15.6	10.6	4.2	3.2
	2%	WB	-2.3	-2.8	-1.4	1.8	5.3	8.8	11.3	10.1	7.9	3.6	0.3	-2.1
		MCDB	0.5	-0.2	1.3	6.1	10.5	13.8	15.9	15.2	14.1	8.1	2.7	1.5
	5%	WB	-3.5	-3.9	-2.4	0.6	4.2	7.3	10.0	9.2	7.0	2.2	-1.2	-3.4
		MCDB	-1.2	-1.6	0.2	4.2	9.0	12.3	14.8	14.3	13.0	6.3	1.2	-0.3
	10%	WB	-4.6	-5.2	-3.5	-0.7	3.0	6.1	9.0	8.3	5.9	1.1	-2.7	-4.6
		MCDB	-2.4	-2.9	-1.2	2.4	7.6	10.4	14.2	13.6	11.7	5.2	-0.5	-2.2
Mean Daily Temperature Range	5% DB	MDBR	5.5	5.7	5.6	5.9	5.8	6.0	7.1	7.0	6.1	5.4	5.0	5.2
		MCDBR	5.8	5.9	6.2	7.4	7.3	7.9	8.4	8.2	7.7	6.7	5.6	6.2
		MCWBR	4.0	3.8	4.5	4.3	3.8	4.4	4.2	4.0	3.8	4.1	4.1	4.4
	5% WB	MCDBR	5.5	5.5	6.1	6.9	6.6	7.4	8.3	8.1	7.5	6.9	5.4	5.9
		MCWBR	4.3	4.4	4.8	4.6	4.0	4.4	4.8	4.5	4.1	4.5	4.4	4.7
Clear-Sky Solar Irradiance	taub	0.206	0.217	0.234	0.270	0.280	0.291	0.285	0.293	0.264	0.237	0.214	0.199	
	taud	2.490	2.492	2.486	2.408	2.513	2.519	2.553	2.539	2.612	2.675	2.586	2.478	
	Ebn at Noon	891	964	995	979	978	965	967	944	947	931	879	850	
	Edh at Noon	52	69	85	104	99	99	95	91	75	58	47	44	
All-Sky Solar Radiation	RadAvg	1.12	2.07	3.24	4.55	5.28	5.58	6.30	5.23	3.68	2.29	1.21	0.88	
	RadStd	0.07	0.14	0.22	0.31	0.45	0.46	0.46	0.37	0.42	0.22	0.08	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page